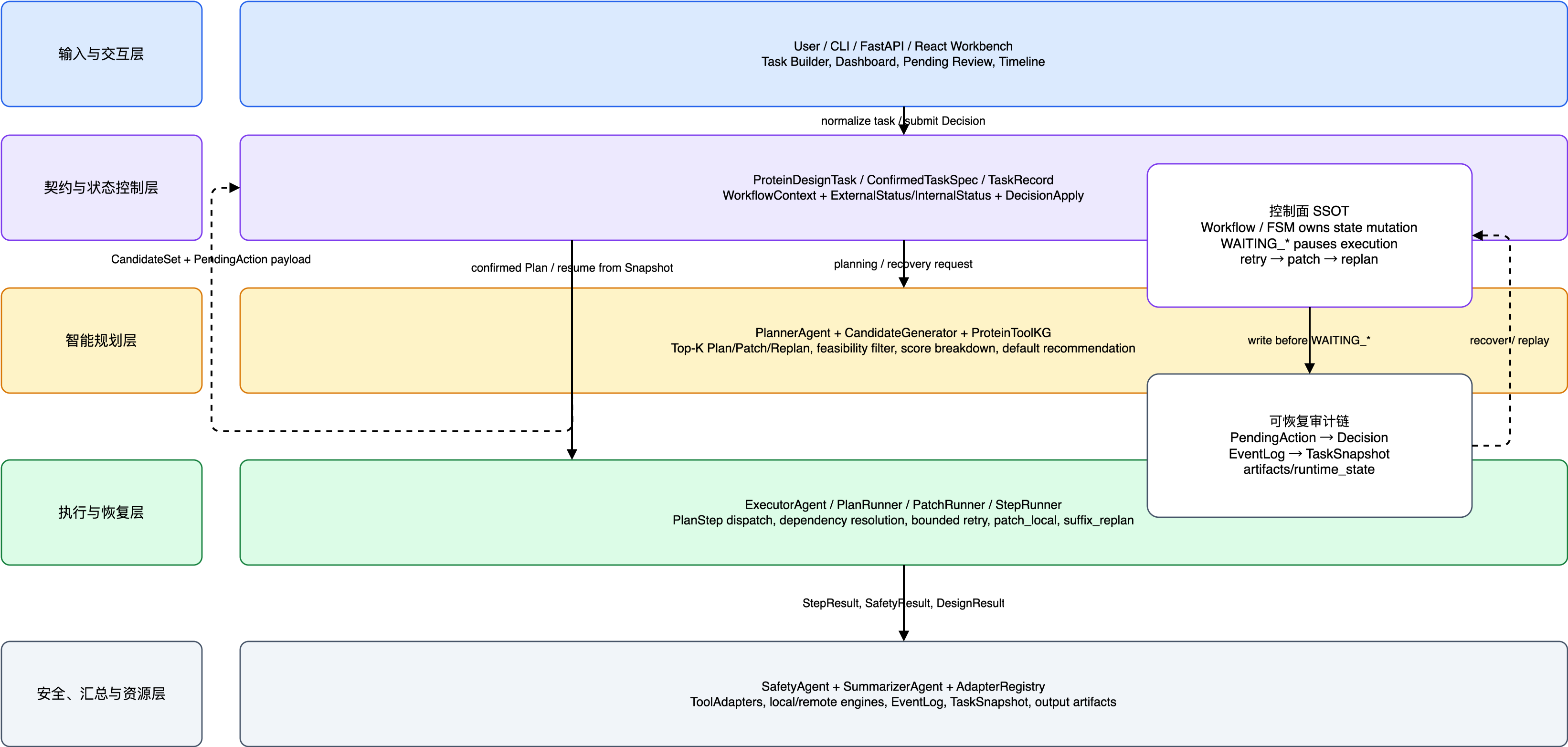


系统架构图：LLM 驱动蛋白质设计多智能体 workflow 系统

五层架构与控制面分离：输入/规划/执行/安全汇总/资源层由 Workflow FSM 与 HITL 审计链统一约束。



Sequence generation
ProtGPT2 / ProteinMPNN

Structure projection
ESMFold / NIM / OpenFold

Quality gate
BioPython QC / DSSP

Objective scoring
objective_ranker / metrics

Visualization/report
Mol* / NGL / HTML

Resource layer details: ProteinToolKG stores capability, I/O schema, cost, safety, compatibility; AdapterRegistry maps tool_id to executable backend.